Benchmarking Machine Learning Models for Road-Level Flood Prediction and Flood-Aware Routing: A Case Study of Da Nang, Vietnam

Anonymous Author(s)
Anonymous Institution
Anonymous City, Country

***Abstract*—During heavy rain, drivers need to know which roads are flooded, yet flood studies in Vietnamese cities map susceptibility over areas, and flood models are compared only by classification scores. It is therefore unknown whether the model that best predicts flooded roads also gives the safest routes. This paper benchmarks six machine learning models from five families—logistic regression, support vector machine, random forest, XGBoost, multilayer perceptron, and the graph neural network GraphSAGE—for road-level flood prediction and for the flood-aware routing built on their predictions, using the extreme rainfall of 14–15 October 2022 in Hai Chau, Da Nang, Vietnam, as a case study. The 2,059 road segments are labeled from a two-dimensional hydraulic simulation of the event. Under spatial block cross-validation, the five nonlinear models performed similarly (area under the precision–recall curve 0.796–0.840, highest for random forest), and the differences were not statistically significant, while random cross-validation overstated AUC-ROC by +0.034 to +0.094, most for tree ensembles. Used as routing costs, the out-of-fold probabilities of the multilayer perceptron reduced flooded distance per trip by 41.2% at a median travel-time increase of 16.5%, so the best classifier was not the best router at a fixed risk weight. Even with full knowledge of the simulated flooding, 94.7% of trips crossed at least one flooded segment. Shapley additive explanations identified depression depth, topographic position index, and distance to the Han River as the main flood drivers.**

***Index Terms*—benchmarking, explainable artificial intelligence, flood-aware routing, graph neural networks, machine learning, road networks, spatial cross-validation, urban flooding**

I. Introduction

Urban flooding disrupts road transport long before it damages buildings. A road becomes unsafe for most vehicles at about 30 cm of standing water [1], and the disruption spreads beyond the flooded links. During Hurricane Harvey, flooding of only 1.3% of the roads in Houston increased network-wide travel times by 8% [2]. For drivers and traffic managers, the useful question during heavy rain is therefore which road segments become impassable, not which areas are susceptible.

Da Nang, Vietnam, illustrates this problem. On the night of 14 October 2022, the city received close to 600 mm of rain; parts of the city stood about 1 m deep, and the Dien Bien Phu underpass flooded completely [3], [4]. Hai Chau, the dense central district along the Han River, is classed among the most flood-susceptible areas of the city [5].

Flood research in Da Nang maps susceptibility with machine learning [5], [6], simulates the sewer network [7], and maps inundation from radar imagery [8], but it reports results for areas, grid cells, or sewer nodes rather than roads. Three gaps remain. First, road-level prediction is rare, and the closest study [9] validated its models with random splits, which overstate accuracy [10]. Second, model comparisons cover mostly tree ensembles [5], [6], [9], [11], so linear, kernel, neural, and graph models are rarely tested on the same data. Third, flood models are judged by classification scores alone. This leaves open the question that this paper asks: *does the model that best predicts flooded roads also give the safest routes?*

To answer it, this paper benchmarks six machine learning models from five families for road-level flood prediction and flood-aware routing in Hai Chau during the 2022 Da Nang flood. Its contributions are as follows:

1) **Leakage-aware classification benchmark:** six models compared under spatial block cross-validation on simulation-derived road labels, with precision–recall, calibration, significance tests, and feature attributions.
2) **Downstream routing benchmark:** the out-of-fold probabilities of every model used as routing costs and compared with shortest-path and oracle baselines.
3) **Classification–routing agreement:** a test of whether the two model rankings agree.
4) **Efficiency and robustness:** the cost of each model, a feature-group ablation, and a sensitivity analysis.

Section II reviews related work, Section III describes the case study, Section IV formulates the problem and the benchmark, Section V reports the results, Section VI discusses them, and Section VII concludes the paper.

II. Related Work

A. Flood Studies in Da Nang

Nguyen et al. [5] compared five tree ensembles across the city (AUC 0.983–0.989) and placed Hai Chau in the high and very-high susceptibility classes, and Hoang et al. [6] trained XGBoost on 1,000 flood points from media reports (accuracy 0.93). Both used random train–test splits, which overstate accuracy when nearby samples are correlated [10]. Doan et al. [7] simulated the 2022 rainfall with MIKE Urban and found that riverside manholes flooded when high river levels blocked the outfalls, and Nguyen and Quach [8] mapped the October

2022 flooding from Sentinel-1 imagery, mostly outside the dense urban core.

B. Road-Level Flood Prediction

Ha et al. [11] mapped flash-flood susceptibility along a highway in Hoa Binh, and Sadler et al. [12] modeled street flood severity from crowdsourced reports. Yuan et al. [9] are closest to this work: random forest predicted flooded roads in Harris County, Texas, from topographic, hydrologic, and precipitation features (AUC 0.86 for Hurricane Harvey), but under random cross-validation. Pan et al. [13] used a Bayesian network to place road flood sensors in New York City; their work observes flooding on segments, whereas this paper predicts it. Graph neural networks such as GraphSAGE [14] can use the connectivity between segments, but they have not been compared with feature-based models for this task.

C. Routing Under Disruption

Le et al. reviewed optimization models for urban route planning [15] and investigated the spatiotemporal impacts of route optimization on road networks [16]. More recently, Le et al. [17] optimized urban bus routes in Da Nang with multi-objective deep reinforcement learning, without hazard costs. Linking segment-level flood probabilities to routing connects these two lines of work.

As Table I shows, no prior study combines road-level segment prediction, spatial block validation, probability calibration, and downstream flood-aware routing evaluation.

III. Case Study

A. Study Area and Historical Flood Event

The study covers the former Hai Chau district of Da Nang, with boundaries as they stood in October 2022, on the west bank of the Han River. The district contains the commercial core of the city, a dense street grid, and the Dien Bien Phu underpass, and it is classed among the most flood-susceptible areas of the city [5].

The historical event studied is the extreme rainfall of 14–15 October 2022. Da Nang received close to 600 mm of rain overnight, parts of the city stood about 1 m deep, and the Dien Bien Phu underpass flooded completely [3], [4]. Manholes along the Han River also overflowed, because high river levels blocked the drainage outfalls [7]. The rain-gauge record used in this study gives 507.3 mm of cumulative rainfall and a peak hourly intensity of 27.1 mm/h. Fig. 1 shows the study area.

B. Road Network and Flood Labels

The drivable network was obtained from OpenStreetMap with OSMnx [18] and split at intersections and at a maximum length of 100 m, giving 1,190 intersections and 2,059 directed road segments.

Flood labels are derived from a two-dimensional HEC-RAS hydraulic simulation of 14–15 October 2022 [need results: name the source of the HEC-RAS model and how it was calibrated or validated]. Points are sampled every 10 m along each segment centerline and matched to the nearest mesh cell,



Fig. 1. Study area, road network, and simulated impassable segments (water depth > 30 cm) in Hai Chau.

and the segment depth is the largest peak water depth over the 37 hourly time steps at these points, so a single low point can make a segment impassable. Following the 30 cm threshold of Pregnolato et al. [1], segments are classed as passable (< 10 cm; 139 segments, 6.8%), flooded but passable (10–30 cm; 1,044 segments, 50.7%), or impassable (> 30 cm; 876 segments, 42.5%).

The target is impassable flooding: $y = 1$ for depth > 30 cm and $y = 0$ otherwise, so flooded-but-passable segments count as non-flooded. Section V-I tests two alternatives: excluding the flooded-but-passable segments, and labeling every segment deeper than 10 cm as flooded.

C. Features

Each segment is described by 12 numerical features and its one-hot-encoded road class. Terrain features are the minimum and mean elevation, mean slope, mean topographic position index (TPI), and maximum depression depth within a 50 m buffer, computed from a 30 m digital elevation model (DEM). The hydrological feature is the distance to the Han River and its channels. Network features are road class, segment length, node degree, edge betweenness centrality, and a flag

TABLE I
SUMMARY OF RELATED FLOOD PREDICTION AND ROUTING STUDIES

Study	Area	Prediction unit	Models compared	Spatial validation	Calibration	Routing tested
Nguyen et al. [5]	Da Nang	Area	Five tree ensembles	×	×	×
Hoang et al. [6]	Da Nang	Point	XGBoost	×	×	×
Ha et al. [11]	Hoa Binh	Road corridor	REPT and hybrid ensembles	×	×	×
Sadler et al. [12]	Norfolk, VA	Street	Poisson regression, RF	×	×	×
Yuan et al. [9]	Harris County	Road	RF, AdaBoost	×	×	×
Pan et al. [13]	New York City	Expressway	Bayesian network	–	–	×
Le et al. [17]	Da Nang	Route	Deep RL	–	–	✓[a]
This paper	Da Nang (Hai Chau)	Road segment	Six models, five families	✓	✓	✓

[a]Routing without flood or hazard costs.

for underpasses, tunnels, and bridges. Rainfall features are the event total and peak hourly intensity at the nearest rain gauge; because one gauge covers the whole district, they are constant across segments and carry no information in this single-event benchmark.

IV. Problem Formulation and Benchmark Design

A. Problem Formulation

The road network is a directed graph $G = (V, E)$, where V is the set of intersections and E is the set of road segments. Each segment $e_i \in E$ has a feature vector $\mathbf{x}_i \in \mathbb{R}^d$, a length ℓ_i, a free-flow travel time t_i, and a flood label $y_i \in \{0, 1\}$. The benchmark has two tasks.

1) Classification task: A model estimates the probability that segment e_i floods:

$$p_i = P(y_i = 1 \mid \mathbf{x}_i). \tag{1}$$

For the graph model, the input also includes the features of neighboring segments, taken from the line graph $L(G)$, whose nodes are the segments of G and whose edges join segments that share an intersection.

2) Routing task: The probabilities are converted into edge costs

$$C_i = t_i\,(1 + \lambda\, p_i), \tag{2}$$

where $\lambda \geq 0$ controls risk aversion. For an origin–destination (OD) pair (o, d) with the set of paths Π_{od}, the flood-aware route is

$$\pi^* = \arg\min_{\pi \in \Pi_{od}} \sum_{e_i \in \pi} C_i. \tag{3}$$

A route π is evaluated by its observed flood exposure and its travel-time increase over the shortest-time route π_0:

$$F(\pi) = \sum_{e_i \in \pi} y_i\, \ell_i, \qquad \Delta T(\pi) = \frac{T(\pi) - T(\pi_0)}{T(\pi_0)}, \tag{4}$$

where $T(\pi) = \sum_{e_i \in \pi} t_i$. The research question is whether the model with the best classification score also gives the lowest F at a given ΔT.

B. Models for Comparison

Six models from five families are compared:

- **Linear:** logistic regression (LR), a transparent baseline.
- **Kernel:** support vector machine (SVM) with a radial basis function kernel [19] and Platt scaling.
- **Tree ensembles:** random forest (RF) [20] and XGBoost [21].
- **Neural:** multilayer perceptron (MLP) with two hidden layers (64 and 32 units) with ReLU activations and dropout (0.2).
- **Graph:** GraphSAGE [14] on the line graph $L(G)$, with two SAGEConv layers (32 hidden channels) and mean neighborhood aggregation.

All models receive the same features and folds and use the fixed random seed 42. Class weights handle class imbalance (a positive-class weight for XGBoost, the MLP, and GraphSAGE). Features are standardized within each training fold for all models except the tree ensembles; GraphSAGE, which is trained transductively, standardizes over all segments. Hyperparameters are fixed rather than tuned: LR uses the liblinear solver, the SVM uses $C = 1$, RF uses 100 trees, XGBoost uses 100 trees of depth 5 with a learning rate of 0.08, and the MLP and GraphSAGE are trained full-batch with Adam (learning rate 0.01, weight decay 10^{-4}) for 80 and 100 epochs.

C. Classification Benchmark

Neighboring segments share terrain and drainage, so random splits leak information between training and test sets [10]. The district is therefore divided into square blocks of 500 m, which are assigned to five folds and held out in turn. For GraphSAGE, held-out segments stay in the graph, but their labels are hidden during training. Random five-fold cross-validation is also run, and the optimism gap of model m is

$$\Delta_m = \mathrm{AUC}_m^{\mathrm{random}} - \mathrm{AUC}_m^{\mathrm{spatial}}. \tag{5}$$

The primary metric is the area under the precision–recall curve (AUC-PR), whose no-skill baseline equals the share of flooded segments (0.425). AUC-ROC, F1, and recall on the flooded class are also reported, and calibration is measured with the Brier score [22], because the probabilities become routing costs. F1 and recall use a threshold of 0.5. Differences

in AUC-PR across folds are tested with the Friedman test and pairwise Wilcoxon signed-rank tests against the best model [23]; with five folds, the smallest attainable two-sided Wilcoxon p-value is 0.0625, so these tests can show trends but not significance at the 0.05 level. Feature effects are explained with TreeSHAP [24] on an XGBoost model fitted to all segments. To compare models, each feature is ranked by a model-specific importance: the absolute coefficient for LR, the impurity importance for RF, the mean absolute SHAP value for XGBoost, and, for the SVM, MLP, and GraphSAGE, the absolute correlation between the feature and the model's out-of-fold probabilities.

D. Routing Benchmark

Every segment receives the probability predicted in the fold where it was held out, so no model routes on labels it was trained on. Free-flow travel times use speeds by road class, from 50 km/h on primary roads to 20 km/h on residential streets. For 150 random OD pairs at least 300 m apart, routes are computed with Dijkstra's algorithm for each model and for two baselines: shortest time ($p_i = 0$) and an oracle ($p_i = y_i$) that knows the simulated labels. Each route is scored by its number and length of flooded segments and its travel-time increase. The main results use $\lambda = 5$; because the models output probabilities on different scales, the same λ makes them detour by different amounts, so λ is also varied over $\{1, 2, 5, 10, 20\}$ on the first 50 OD pairs to compare models at equal travel-time cost. Finally, the model rankings by classification and by routing are compared.

E. Ablation, Sensitivity, and Efficiency

The feature-group ablation removes the terrain, hydrological, rainfall, or network features in turn for XGBoost and GraphSAGE. For GraphSAGE, the number of layers is varied, and a version without graph edges isolates the effect of connectivity. Sensitivity is tested with XGBoost for the block size (250, 500, and 1000 m) and for the labeling of flooded-but-passable segments. Efficiency is measured by the number of parameters, the training time, the inference time for the district, and the routing time per OD pair, on an AMD Ryzen 7 processor with 16 GB RAM running PyTorch 2.14, XGBoost 3.2, and PyTorch Geometric 2.8. Results are reported as the mean $\pm$ standard deviation over the five spatial folds for a single seed.

V. Results

A. Flood Labels

Of the 2,059 segments, 876 (42.5%) are impassable in the simulation ($y = 1$), 1,044 (50.7%) are flooded but passable, and 139 (6.8%) are passable. Only 6.8% of the network stays below 10 cm, so the event affected almost the whole district. The impassable segments cluster along the low-lying bank of the Han River and around natural depressions, including the Dien Bien Phu underpass.

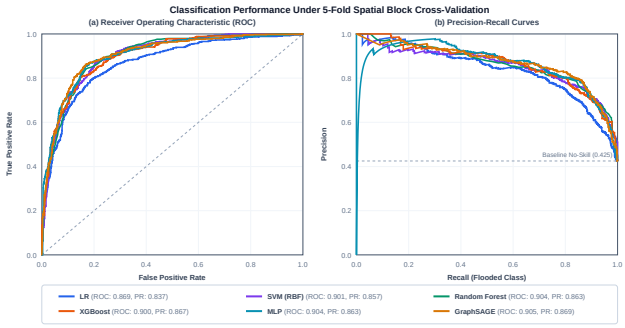


Fig. 2. ROC (left) and precision–recall (right) curves under spatial block cross-validation. The dashed line in the right panel is the no-skill baseline (0.425). Legend values are computed on pooled out-of-fold predictions, so they differ slightly from the fold means in Table II.

B. Classification Benchmark

Table II compares the six models, and Fig. 2 shows their ROC and precision–recall curves under spatial cross-validation. Random forest achieved the highest AUC-PR (0.840 ± 0.077), followed by the MLP (0.833 ± 0.067), XGBoost (0.831 ± 0.081), and GraphSAGE (0.823 ± 0.083); logistic regression was lowest (0.784 ± 0.111). These differences are small compared with the variation between folds. The Friedman test found no significant difference among the six models ($\chi^2 = 7.86$, $p = 0.164$), and pairwise Wilcoxon tests against random forest gave $p = 0.0625$ for LR, the smallest value possible with five folds, and $p \geq 0.31$ for the other models. GraphSAGE reached the highest recall on impassable segments (0.852 ± 0.043) and F1 (0.791 ± 0.057), and random forest the lowest Brier score (0.127 ± 0.012).

C. Optimism of Random Cross-Validation

Fig. 3 compares AUC-ROC under random and spatial cross-validation. Every model scored higher under random cross-validation, with optimism gaps Δ_m from $+0.034$ (LR) to $+0.094$ (random forest). The tree ensembles were inflated most (random forest $+0.094$, XGBoost $+0.085$), followed by GraphSAGE ($+0.077$), the MLP ($+0.054$), and the SVM ($+0.047$). Under random cross-validation, random forest would appear near-perfect (AUC-ROC 0.992); under spatial cross-validation, all models except LR lie between 0.889 and 0.897.

D. Risk Map and Feature Effects

Fig. 4 maps the out-of-fold flood probability of each segment for random forest, the best classifier. [need results: verify the share of segments with $p \geq 0.66$ (currently stated as 36.4%) from the notebook] of the segments fall in the high-risk class, concentrated in the low-lying wards along the Han River and near the Dien Bien Phu underpass. Fig. 5 shows the TreeSHAP values of XGBoost. The top features are depression depth (mean $|\text{SHAP}| = 1.592$), TPI (0.992), distance to the Han River (0.499), and mean elevation (0.486); deeper depressions and lower ground raise the flood probability. Fig. 6

TABLE II
CLASSIFICATION BENCHMARK UNDER SPATIAL BLOCK CROSS-VALIDATION (MEAN ± STANDARD DEVIATION OVER FIVE FOLDS)

Model	Family	Spatial CV					Random CV
		AUC-PR	AUC-ROC	F1	Recall (flooded)	Brier score ↓	AUC-ROC
Logistic regression	Linear	0.784 ± 0.111	0.862 ± 0.023	0.739 ± 0.086	0.757 ± 0.108	0.148 ± 0.013	0.897 ± 0.017
SVM (RBF)	Kernel	0.796 ± 0.095	0.889 ± 0.019	0.745 ± 0.097	0.755 ± 0.102	0.129 ± 0.013	0.936 ± 0.009
Random forest	Tree ensemble	**0.840 ± 0.077**	**0.897 ± 0.023**	0.757 ± 0.110	0.747 ± 0.139	**0.127 ± 0.012**	0.992 ± 0.005
XGBoost	Tree ensemble	0.831 ± 0.081	0.893 ± 0.025	0.747 ± 0.092	0.749 ± 0.145	0.136 ± 0.017	0.978 ± 0.007
MLP	Neural	0.833 ± 0.067	0.893 ± 0.025	0.784 ± 0.065	0.816 ± 0.097	0.132 ± 0.018	0.948 ± 0.011
GraphSAGE	Graph	0.823 ± 0.083	0.892 ± 0.032	**0.791 ± 0.057**	**0.852 ± 0.043**	0.139 ± 0.030	0.969 ± 0.011

Best spatial-CV value per column in bold. ↓: lower is better. No-skill AUC-PR: 0.425.

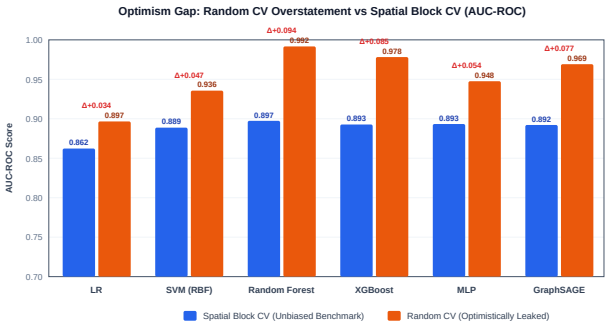


Fig. 3. AUC-ROC of each model under spatial block and random cross-validation. Labels give the optimism gap Δ_m.

compares the feature rankings of the six models. All models rank terrain features highest: depression depth is first for four models and second for the other two, and TPI is first for LR and random forest. Network features such as betweenness and segment length rank low for every model, including GraphSAGE.

E. Feature-Group Ablation

Table III reports AUC-PR when each feature group is removed. Terrain features matter most: removing them lowers AUC-PR from 0.831 to 0.575 for XGBoost and from 0.823 to 0.484 for GraphSAGE. Removing the distance to the river costs 0.019 for XGBoost and 0.070 for GraphSAGE. Removing the rainfall or network features leaves XGBoost unchanged, as expected for the constant rainfall features. For GraphSAGE, the same constant rainfall features change AUC-PR by -0.019, which shows that its run-to-run variation is about 0.02. The graph ablations (-0.016 without edges, $+0.001$ with one layer) fall within this variation, so the benchmark gives no evidence that road connectivity improves prediction; only a third layer clearly hurts (-0.070).

F. Downstream Routing

Table IV compares the routing results of the six models with the two baselines, and Fig. 7 plots flood exposure against travel-time increase. At $\lambda = 5$, the MLP gave the fewest flooded segments (10.35 per trip) and the shortest flooded length (625.8 m), 41.2% less than the shortest-time route, at

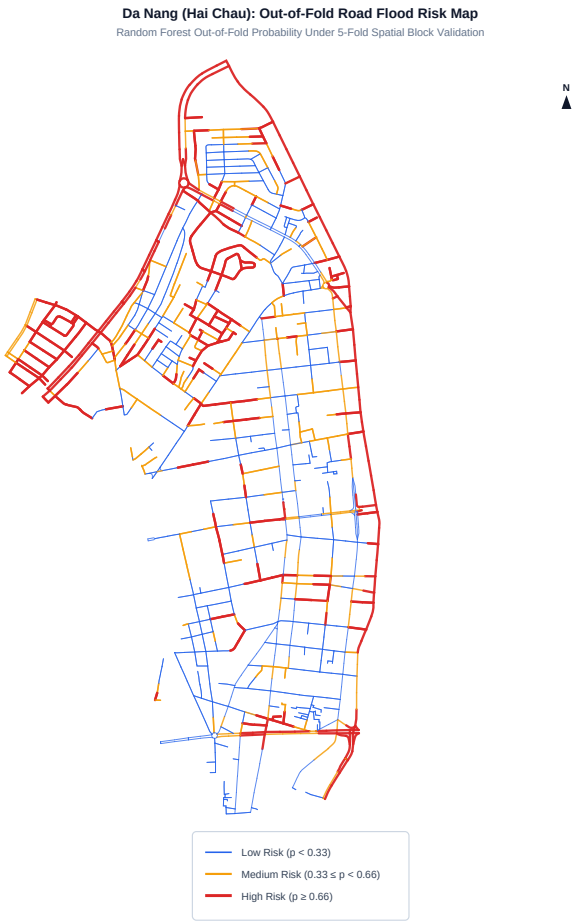


Fig. 4. Out-of-fold flood probability per road segment from random forest, in three risk classes.

a median travel-time increase of 16.5%. XGBoost followed (639.8 m, +13.5%), and random forest, the best classifier, ranked third (683.6 m) with the smallest detour among the nonlinear models (+7.6%). The routing ranking therefore differs from the AUC-PR ranking at the top, although LR is last on both. Part of the difference comes from detour size: the models' probabilities lead to different detours at the same λ. Fig. 8 compares the models at equal travel-time cost as λ varies; there the curves of the nonlinear models lie close

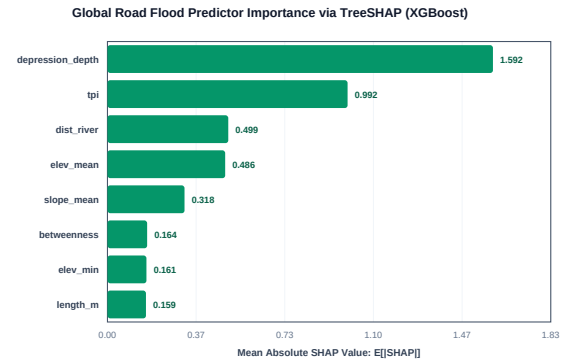


Fig. 5. Mean absolute TreeSHAP values of XGBoost fitted to all segments.

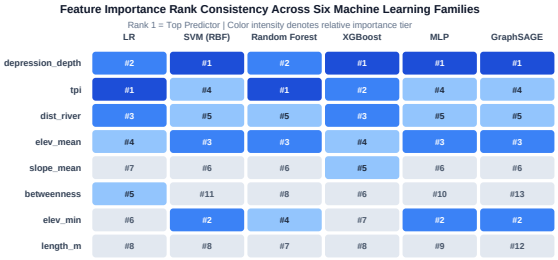


Fig. 6. Rank of each feature by model-specific importance across the six models (1 = most important).

together, with XGBoost lowest over most of the range. Even the oracle route still crossed at least one flooded segment in 94.7% of the OD pairs (8.03 segments and 485.6 m per trip), because 42.5% of the network is impassable and long detours are penalized.

G. Representative Route

Fig. 9 shows one OD pair crossing the central corridor, chosen as a case in which the shortest-time route crosses at least 400 m of flooding and the oracle avoids it completely; Table V lists the metrics. The shortest-time route crosses 14 flooded segments (951.0 m). The MLP route reduces this to 2 segments (108.4 m) for 1.85 extra minutes, and the XGBoost route to 6 segments (344.2 m) for 1.24 extra minutes.

H. Computational Efficiency

Table VI reports the size and run time of each model. All models trained in under 0.5 s and predicted the whole district in under 90 ms. LR was the fastest to train (0.01 s) and predict (2.4 ms), and the SVM the slowest to predict (88.8 ms). Routing took 2.3 ms per OD pair.

I. Sensitivity Analysis

Fig. 10 and Table VII show how XGBoost's results change with the block size and the labeling. AUC-PR varied between 0.831 and 0.883 across block sizes of 250 to 1000 m, with the

TABLE III
FEATURE-GROUP AND GRAPH ABLATION (AUC-PR UNDER SPATIAL CV)

Setting	XGBoost	GraphSAGE	ΔAUC-PR
All features (baseline)	0.831	0.823	–
− Terrain	0.575	0.484	−0.339
− Hydrological	0.812	0.753	−0.070
− Rainfall (constant)	0.831	0.804	−0.019
− Network	0.831	0.845	+0.022
GraphSAGE, no graph edges	–	0.807	−0.016
GraphSAGE, 1 layer	–	0.824	+0.001
GraphSAGE, 3 layers	–	0.753	−0.070

Δ: change for GraphSAGE relative to all-feature baseline.

TABLE IV
ROUTING RESULTS OVER 150 OD PAIRS ($\lambda = 5.0$)

Strategy	Flooded seg. per trip	Flooded length (m)	Time increase (median, %)
Shortest time	16.76	1063.6	0.0
LR	12.87	780.7	4.1
SVM (RBF)	11.25	692.6	11.6
Random forest	11.15	683.6	7.6
XGBoost	10.47	639.8	13.5
MLP	**10.35**	**625.8**	16.5
GraphSAGE	11.20	689.5	18.9
Oracle (simulated)	8.03	485.6	21.9

Oracle still crosses flooding in 94.7% of OD pairs.

highest value for the smallest blocks, which leave more neighbors of each test segment in the training set. The alternative labelings give AUC-PR values near 0.99, but these are close to their much higher no-skill baselines: excluding flooded-but-passable segments leaves 86.3% of the remaining segments flooded, and labeling every segment deeper than 10 cm as flooded makes 93.2% flooded. Their AUC-ROC values (0.940 and 0.853) show that the task is easier when the 10–30 cm segments are removed and harder when they count as flooded.

VI. DISCUSSION

A. Classification Scores and Routing Decisions

The best classifier was not the best router. Random forest had the highest AUC-PR, yet at $\lambda = 5$ the MLP and XGBoost routes crossed less flooding. Two effects explain this. First, the five nonlinear models are statistically indistinguishable in AUC-PR, so small differences in score need not carry over to routes. Second, a fixed λ converts each model's probabilities into costs on a different scale: models with higher probabilities on flooded segments detour more, which lowers exposure but raises travel time. When the models are compared at equal travel-time cost (Fig. 8), their curves lie close together. For practitioners, models for flood-aware routing should therefore be compared on exposure–time curves, or at least by recall and calibration, rather than by AUC at a single operating point.

B. Validation Practice

Random cross-validation inflated AUC-ROC for every model, by up to 0.094, and made random forest appear near-perfect (0.992). The inflation was largest for the tree ensem-

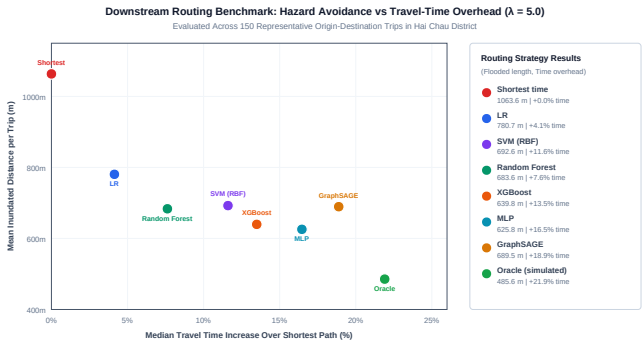


Fig. 7. Flood exposure per trip versus median travel-time increase for each strategy at $\lambda = 5.0$.

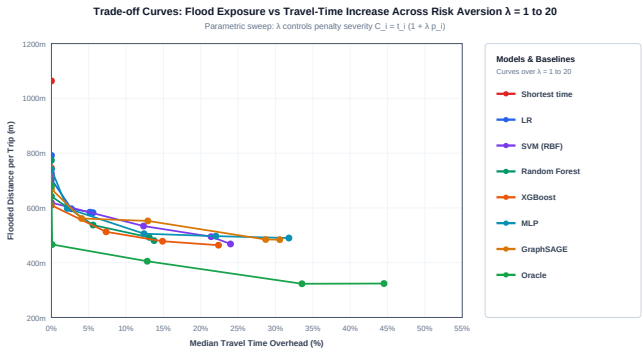


Fig. 8. Trade-off between flood exposure and travel-time increase as risk aversion λ varies from 1 to 20 (first 50 OD pairs).

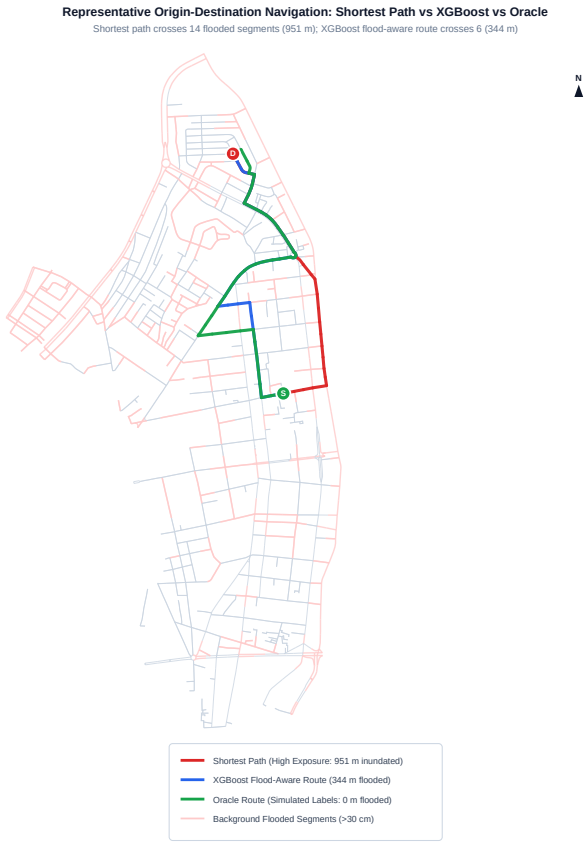


Fig. 9. Routes for a representative OD pair: shortest time (red), XGBoost flood-aware route (blue), and oracle on simulated labels (green).

bles, which can memorize local terrain values that neighboring segments share. Studies that report very high accuracy for urban flood susceptibility with random splits [5], [6], [9] should therefore be read with caution, and spatial blocking should be the default for road-level benchmarks.

C. Flood Drivers

All models ranked terrain first, led by depression depth and TPI, followed by distance to the Han River. This agrees with the hydraulic mechanism in Da Nang, where high river levels blocked the outfalls and water ponded in low-lying streets [7]. However, the labels come from a hydraulic simulation driven by terrain, so part of this agreement is expected by construction: the models partly learn to reproduce the simulation from the same terrain. Network features added little, and road connectivity did not measurably improve GraphSAGE.

D. Limitations

Six limitations apply. First, the labels are simulated rather than observed, so the models predict where the HEC-RAS model floods, and their accuracy against real flooding depends on how well that model was validated. Second, the study uses one event in one district, so the rainfall features are constant and the models identify flood-prone segments rather than forecasting floods for other storms. Third, a 30 m DEM

TABLE V
METRICS OF THE REPRESENTATIVE ROUTE

Strategy	Travel time (min)	Flooded segments	Flooded length (m)
Shortest time	3.08	14	951.0
XGBoost	4.32	6	344.2
MLP	4.93	2	108.4
Oracle (simulated)	5.08	0	0.0

may miss street-scale depressions. Fourth, hyperparameters were fixed rather than tuned, and results come from a single seed, so small differences between models should not be over-interpreted. Fifth, the routing compares models at one λ in Table IV, which mixes prediction quality with probability scale. Sixth, 94.7% of trips cross flooding even with perfect knowledge, so route choice alone cannot keep vehicles out of the water during an event of this size.

VII. Conclusion

This paper benchmarked six machine learning models from five families for road-level flood prediction and flood-aware routing during the 2022 Da Nang flood, using labels from a hydraulic simulation. Under spatial cross-validation, the five nonlinear models performed similarly (AUC-PR 0.796–0.840, highest for random forest), and random cross-validation

TABLE VI
MODEL SIZE AND RUN TIME

Model	Parameters	Training time (s)	Inference, district (ms)	AUC-PR
LR	19	0.01	2.4	0.784
SVM (RBF)	37,062[a]	0.30	88.8	0.796
RF	50,000[a]	0.25	43.3	**0.840**
XGBoost	3,200[a]	0.11	5.5	0.831
MLP	3,329	0.32	12.3	0.833
GraphSAGE	3,297	0.43	8.8	0.823

[a]Approximate size. Routing time per OD pair: 2.3 ms.

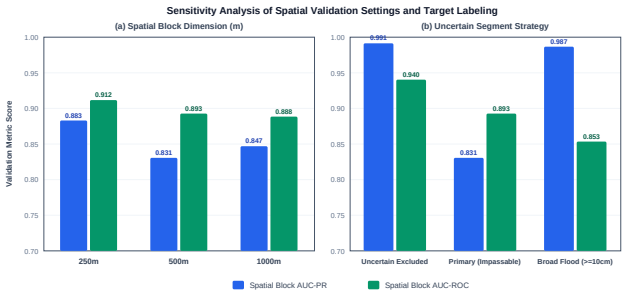


Fig. 10. Sensitivity of the results to (a) spatial block size and (b) handling of uncertain segments.

overstated AUC-ROC by up to 0.094. As routing costs, the MLP's probabilities reduced flooded distance by 41.2% for a 16.5% increase in travel time, so the best classifier was not the best router at a fixed risk weight, and even an oracle could not avoid flooding on 94.7% of trips. Terrain, led by depression depth, drove the predictions of every model.

Future work will validate the labels against observed flood reports, add further events, tune hyperparameters over several seeds, use timestamped depth records for short-term prediction, and add risk costs to reinforcement-learning routing [17]. The risk maps could also provide prior flood probabilities for placing road sensors or cameras in Da Nang, following [13].

TABLE VII
SENSITIVITY OF XGBOOST TO VALIDATION AND LABELING SETTINGS

Setting	AUC-PR	No-skill AUC-PR	AUC-ROC
Block size 250 m	0.883	0.425	0.912
Block size 500 m (main)	0.831	0.425	0.893
Block size 1000 m	0.847	0.425	0.888
10–30 cm segments excluded	0.991	0.863	0.940
10–30 cm as non-flooded (main)	0.831	0.425	0.893
10–30 cm as flooded ($\geq$10 cm)	0.987	0.932	0.853

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